

Prompt:

Write a Python program using a while loop to calculate the area of a Square, Circle, or Rectangle. Display the three options to the user and allow the user to choose only one option. Based on the selected option, get the required input values and calculate and display the area. If the user enters an invalid option, display an error message and ask them to choose again. Use break to exit the loop after a valid choice.

Program:

```
import math
```

```
while True:
```

```
    print("\n1. Square")
```

```
    print("2. Circle")
```

```
    print("3. Rectangle")
```

```
    choice = int(input("Enter your choice: "))
```

```
    if choice == 1:
```

```
        side = float(input("Enter the side: "))
```

```
        area = side * side
```

```
        print("Area of square =", area)
```

```
        break
```

```
    elif choice == 2:
```

```
        radius = float(input("Enter the radius: "))
```

```
area = math.pi * radius * radius
```

```
print("Area of circle =", area)
```

```
break
```

```
elif choice == 3:
```

```
length = float(input("Enter the length: "))
```

```
breadth = float(input("Enter the breadth: "))
```

```
area = length * breadth
```

```
print("Area of rectangle =", area)
```

```
break
```

```
else:
```

```
print("Invalid choice! Please choose 1, 2, or 3.")
```

Sample Outputs:

1. Square

2. Circle

3. Rectangle

Enter your choice: 1

Enter the side: 5

Area of square = 25.0

1. Square
2. Circle
3. Rectangle

Enter your choice: 2

Enter the radius: 7

Area of circle = 153.94

1. Square
2. Circle
3. Rectangle

Enter your choice: 3

Enter the length: 10

Enter the breadth: 5

Area of rectangle = 50.0